

Question	Marking details	Marks available					
		AO1	AO2	AO3	Total	Maths	Prac
7	• Extract mRNA from the bacterium (which is synthesising the proteins). • Use reverse transcriptase to synthesise a {single strand of DNA/ cDNA} using the mRNA (as a template). • Use DNA polymerase to produce double stranded DNA. • Sticky ends are added to the end of each DNA strand of the DNA. • The DNA is copied many times using the PCR. • Plasmids from bacteria are isolated • Restriction {endonuclease/ enzymes} is used to open the plasmid (at a specific point) • producing sticky ends complementary to the DNA. • DNA ligase is used to bind the DNA into the plasmid. • Antibiotic resistance genes / markers/ LacZ are used to select bacteria containing the recombinant plasmids. Concerns: • dispersal of pollen from engineered crops to wild relatives/ contamination of organic crops • unknown effects of eating new protein which is toxic to insects/ health concerns from eating GM crop • a reduction in biodiversity e.g. beneficial insects killed/ affect on food chains • Could result in resistance to protein in insects/ selection pressure						

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	7-9 marks Indicative content of this level is... Detailed explanation of all three areas in the indicative content. <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is... Detailed explanation of 2 areas of the indicative content, or partial description of 3 areas. <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is... Good explanation of one area of the indicative content or partial description of 2/3 areas. <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i>						

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				0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question7 total	7	2		9	0	0